



EDA ASSIGNMENT – NYC TAXI DATASET (CUSTOMIZED)



Project Title

NYC Taxi Trip Analysis – Data Cleaning & Insight Generation



Objective

You are working as a **Data Analyst** in a ride-hailing company.

The company wants to understand:

- Trip behavior
 - Pricing patterns
 - Demand trends
 - Operational inefficiencies
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Tools (MANDATORY)

-  NumPy → Numerical analysis
 -  Pandas → Data cleaning & manipulation
 -  Matplotlib → Basic visualization
 -  Seaborn → Advanced visualization
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Dataset Columns (Given)

Data Set Link: [📄 yellow_tripdata_2020-06](#)

IMP: Randomly Collect 45k Rows And work With Randomised Data

- VendorID

- tpep_pickup_datetime
 - tpep_dropoff_datetime
 - passenger_count
 - trip_distance
 - RatecodeID
 - store_and_fwd_flag
 - PULocationID
 - DOLocationID
 - payment_type
 - fare_amount
 - extra
 - mta_tax
 - tip_amount
 - tolls_amount
 - improvement_surcharge
 - total_amount
 - congestion_surcharge
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SECTION 1: Data Cleaning & Preparation (Step-by-Step)



Step 1: Initial Exploration

- Understand dataset shape
 - Identify column types
 - Separate:
 - Numerical columns
 - Categorical columns
 - Datetime columns
-



Step 2: NumPy-Based Analysis

Perform:

- Mean, median, std of:

- trip_distance
- fare_amount
- total_amount



Goal:

Detect abnormal values



Step 3: Data Type Corrections

- Convert:
 - pickup & dropoff → datetime
 - numeric columns → float/int
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Step 4: Missing Value Handling

Check missing values in:

- passenger_count
- trip_distance
- payment_type
- tip_amount



Decide:

- Fill or drop
-



Step 5: Remove Invalid Records (VERY IMPORTANT)

Remove rows where:

- passenger_count = 0
 - trip_distance ≤ 0
 - fare_amount ≤ 0
 - total_amount ≤ 0
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Step 6: Duplicate Handling

- Check duplicate rows
 - Remove if found
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Step 7: Categorical Cleaning

Standardize:

- payment_type
- store_and_fwd_flag

👉 Ensure:

- Consistent values
-

Step 8: Datetime Feature Engineering

From pickup time, create:

- pickup_hour
 - pickup_day
 - day_of_week
 - month
-

Step 9: Trip Duration

Create:

- trip_duration = dropoff - pickup

👉 Remove:

- Negative durations
 - Unrealistic durations
-



Step 10: Outlier Detection

Identify outliers in:

- trip_distance
- fare_amount
- total_amount
- tip_amount

👉 Use:

- Boxplots
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Step 11: Feature Engineering

Create:

- fare_per_km
 - tip_percentage
 - trip_speed
-



Step 12: Final Validation

Ensure:

- Clean dataset
 - No invalid values
 - Ready for analysis
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SECTION 2: EDA (Step-by-Step)



Step 1: Overall Statistics

- Total trips
 - Average fare
 - Average distance
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Step 2: Univariate Analysis (Matplotlib)

- Distribution of:
 - trip_distance
 - fare_amount
 - passenger_count
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Step 3: Time-Based Analysis (Seaborn)

- Trips by hour
- Trips by day

👉 Identify:

- Peak hours
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Step 4: Distance vs Fare

- Analyze relationship

👉 Insight:

- Pricing consistency
-



Step 5: Tip Analysis

- Tip vs fare
- Tip vs distance

👉 Insight:

- Customer behavior
-



Step 6: Duration Analysis

- Duration vs distance



Insight:

- Traffic conditions
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Step 7: Passenger Analysis

- Trips vs passenger_count
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Step 8: Location Analysis

- PULocation vs DOLocation



Identify:

- Popular routes
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Step 9: Correlation Analysis

- Heatmap between:
 - fare
 - distance
 - duration
 - tips
-
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SECTION 3: Insight Questions (Medium–Hard)



Basic Insights

1. What is the average trip distance?
 2. What is the average fare amount?
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Trip Behavior

3. What is the most common passenger count?
 4. What is the most frequent trip distance range?
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Time Insights

5. At what hour are most trips taken?
 6. Which day has highest demand?
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Pricing Insights

7. Does fare increase with distance?
 8. Are there pricing anomalies?
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Tip Insights

9. Do longer trips give higher tips?
 10. Which trips receive no tips?
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Operational Insights

11. Identify peak hours
 12. Compare weekday vs weekend demand
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Outlier Insights

13. Identify extremely high fares
 14. Identify unrealistic trips
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Advanced Insights

15. Identify inefficient trips (long duration, low distance)
 16. Identify abnormal pricing patterns
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Visualization Tasks

17. Trips per hour plot
 18. Fare vs distance scatter
 19. Boxplot for outliers
 20. Correlation heatmap
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SECTION 4: Deliverables

Students must submit:

- ✓ Cleaned dataset
- ✓ Notebook (NumPy + Pandas + Matplotlib + Seaborn)
- ✓ Visualizations
- ✓ Insight summary



Important Instructions

- Do not skip cleaning
 - Handle invalid data carefully
 - Focus on logic
 - Explain insights clearly
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Final Note

This is a **real-world dataset**.

Focus on:

- Data cleaning
- Pattern recognition
- Insight generation